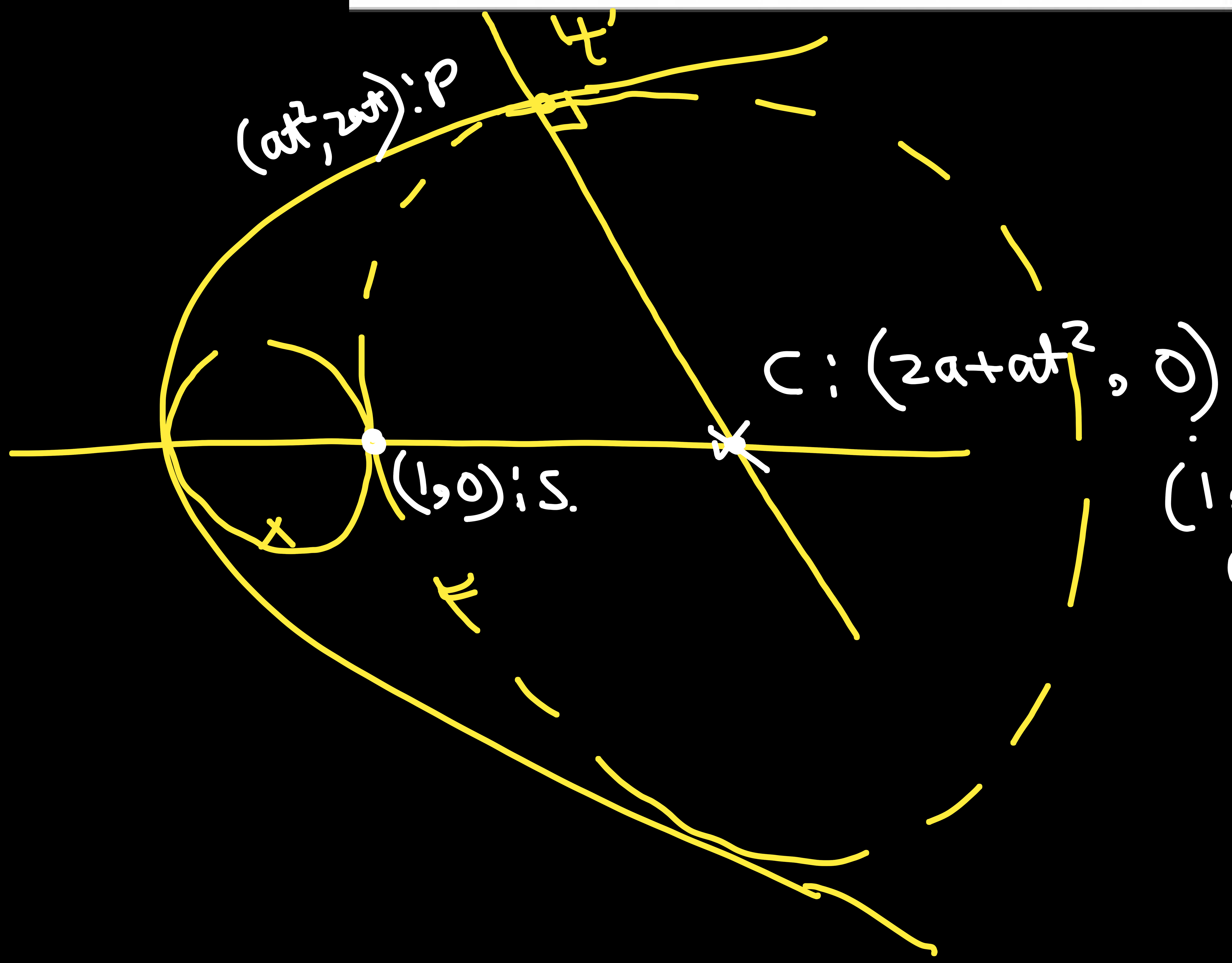
One

Radius of the largest circle which passes through the focus of the parabola $y^2 = 4x$ and contained in it, is

(A) 8 (B) 4 (C) 2 (D) 5



radius of circle is $PC = \sqrt{(2a)^2 + (2at)^2}$.

$(x - 2a - at^2)^2 + y^2 = (2a)^2 + (2at)^2$.

(1,0), $a=1$

$(1 - 2 - t^2)^2 + 0 = 4 + 4t^2$.

$(1+t^2)^{\cancel{2}} = 4\cancel{(1+t^2)} \Rightarrow 1 + t^2 = 4$

$t^2 = 3$.

$r = \sqrt{4(1+t^2)} = \sqrt{4 \times 4} = \boxed{4}$

B

Que

The locus of the vertices of the parabolas represented by the equation

$$y=\frac{a^3x^2}{3}+\frac{a^2x}{2}-2a$$

(a is a non-zero parameter) is

A) a circle

B) a parabola

C) a straight line

D) $xy=c^2$ form where 'c' is constant

Sol

$$y=\left(\frac{a^3}{3}\right)x^2+\left(\frac{a^2}{2}\right)x-2a$$

$$A=\frac{a^3}{3}$$

$$B=a^2/2$$

$$C=-2a$$

$$D=\frac{a^4}{4}+4\cdot\frac{a^3}{3}\cdot 2a$$

$$=a^4\left(\frac{1}{4}+\frac{8}{3}\right)=\frac{35}{12}a^4.$$

$$V:\left(-\frac{B}{2A},-\frac{D}{4A}\right)$$

$$\hookrightarrow x \quad \hookrightarrow y$$

$$x=-\frac{a^2}{2}\cdot\frac{3}{2a^3}=-\frac{3}{4a}.$$

$$y=-\frac{35}{\cancel{12}_4}\cdot\frac{a^4}{4a^3}\times\cancel{3}=-\frac{35}{16}\cdot a.$$

$$\boxed{xy=\frac{35\times3}{64}=\frac{105}{64}}$$

D

Qu Prove that the locus of the centre of the circle, which passes through the vertex of a parabola and through its intersections with a normal chord, is the parabola $2y^2 = ax - a^2$. Equation of given parabola is $y^2 = 4ax$.

Sol

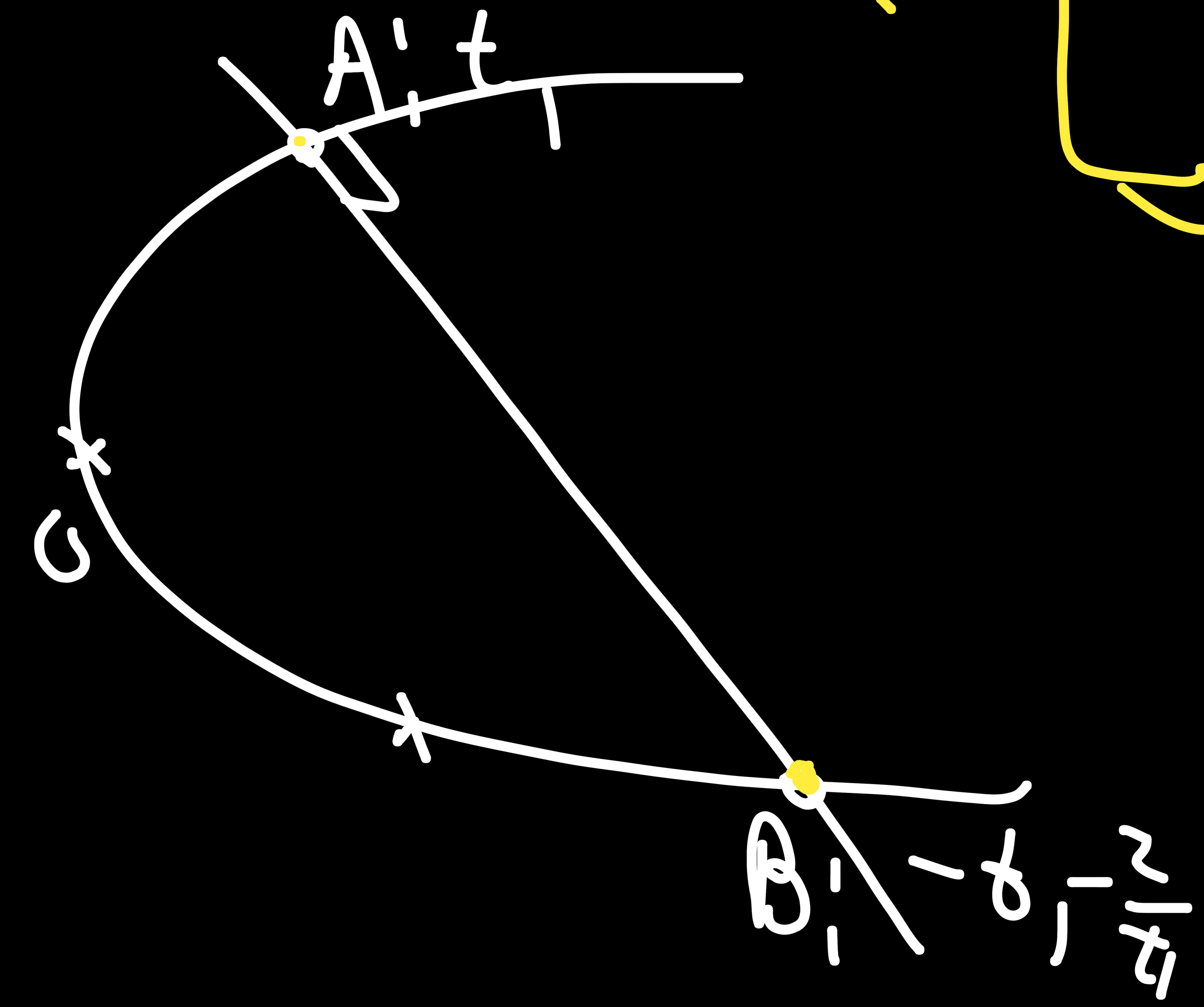
Let circle is $x^2+y^2+2gx+2fy+c=0$

$(at^2, 2at) \rightarrow a^2t^4+4a^2t^2+2gat^2+4aft+c=0 \rightarrow 0,\ t_1,\ -t_1-\frac{2}{t_1},\ \frac{2}{t_1}$

$\Downarrow$ $c=0$.

$\rightarrow a^2t^4+(4a^2+2ga)t^2+4aft=0$

Let normal chord is AB with



$\div t$

$\rightarrow at^3+2(2a+g)t+4f=0 \rightarrow t_1,\ -t_1-\frac{2}{t_1},\ \frac{2}{t_1}$

$\Sigma t_1t_2 = \frac{2(2a+g)}{a} = \cancel{2} - t_1^2 - \cancel{2} - 2 - \frac{4}{t_1^2}$.

$2+\frac{4a+2g}{a} = -\left(t_1^2+\frac{4}{t_1^2}\right)$ —— (1)

$$\text{Product} = -\frac{4f}{a} = \cancel{t_1}\left(-t_1 - \frac{2}{t_1}\right)\cdot\left(\frac{2}{\cancel{t_1}}\right)$$

$$\frac{2f}{a} = -\left(t_1 + \frac{2}{t_1}\right)$$

$$\Rightarrow \frac{4f^2}{a^2} = \underbrace{t_1^2 + \frac{4}{t_1^2}} + 4 = -\left(2 + \frac{4a + 2g}{a}\right) + 4$$

$$-g \longrightarrow x$$
$$-f \longrightarrow y$$

$$\frac{4y^2}{a^2} = 2 - \frac{4a - 2x}{a} = -2 + \frac{2x}{a}.$$

$$\boxed{2y^2 = ax - a^2.}$$

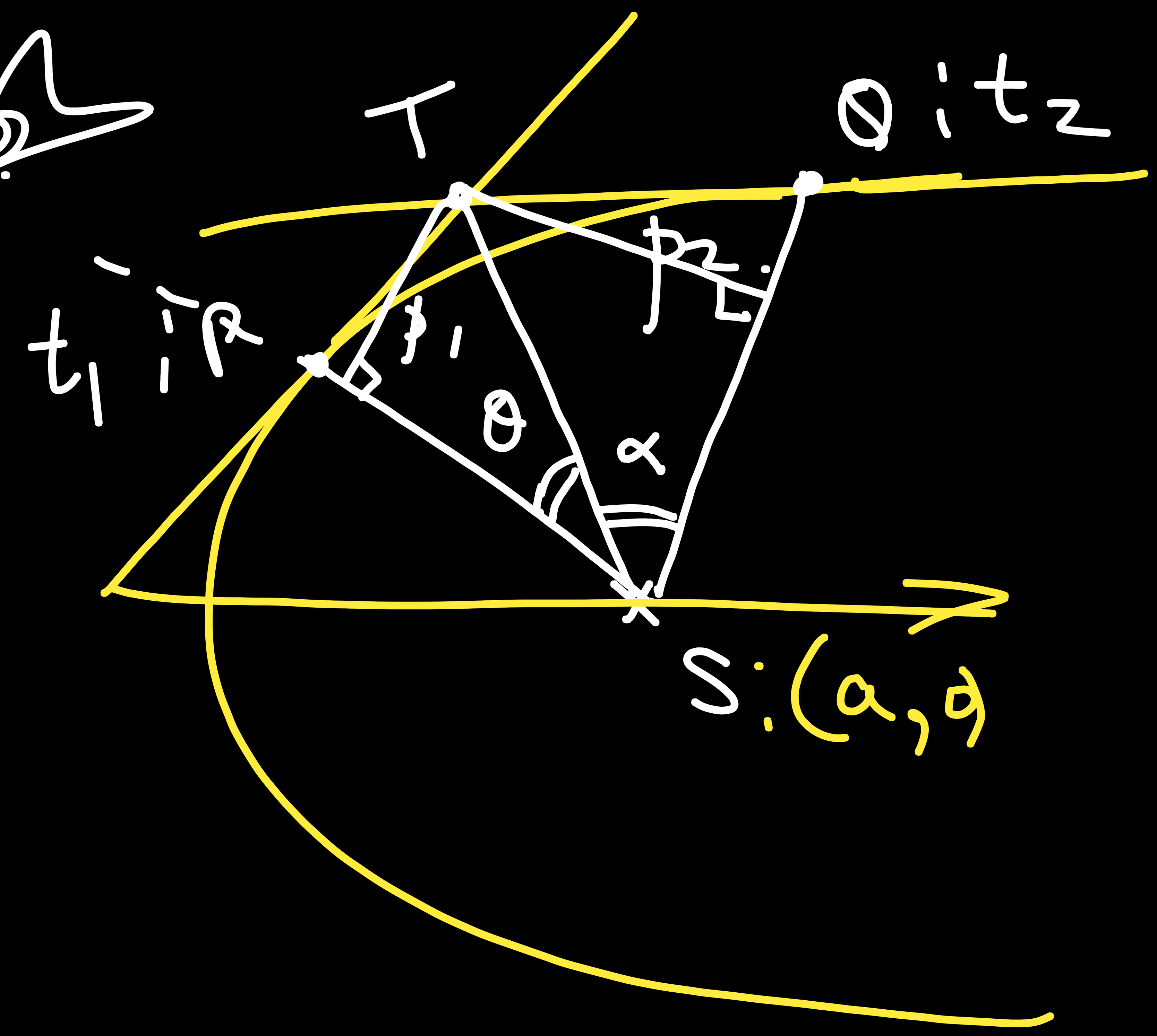
Ans

Qu

If the tangents at P and Q meet in T, then Prove that :-

- TP and TQ subtend equal angles at the focus S.
- $ST^2 = SP \cdot SQ$ &
- The triangles SPT and STQ are similar.

Sol



T : $(at_1t_2, a(t_1+t_2))$

To prove $\theta = \alpha$

we can show that ST is angular bisector of $\angle PSQ$

$\Rightarrow$ T is equidistant from lines PS & SQ.

$$m_{SP} = \frac{0-2at_1}{a-at_1^2} = \frac{2t_1}{t_1^2-1}$$

$$SP \equiv y - 0 = \frac{2t_1}{t_1^2-1}(x-a)$$

$$2t_1x - (t_1^2-1)y - 2at_1 = 0.$$

length of $\perp$'r from T

$$2t_1 x - (t_1^2 - 1)y - 2at_1 = 0$$

$T_1'\left(at_1 t_2,\ a(t_1+t_2)\right)$

$$p_1 = \frac{\left|2t_1\, a t_1 t_2 - (t_1^2-1)\cdot a(t_1+t_2) - 2at_1\right|}{\sqrt{4t_1^2 + (t_1^2-1)^2}}$$

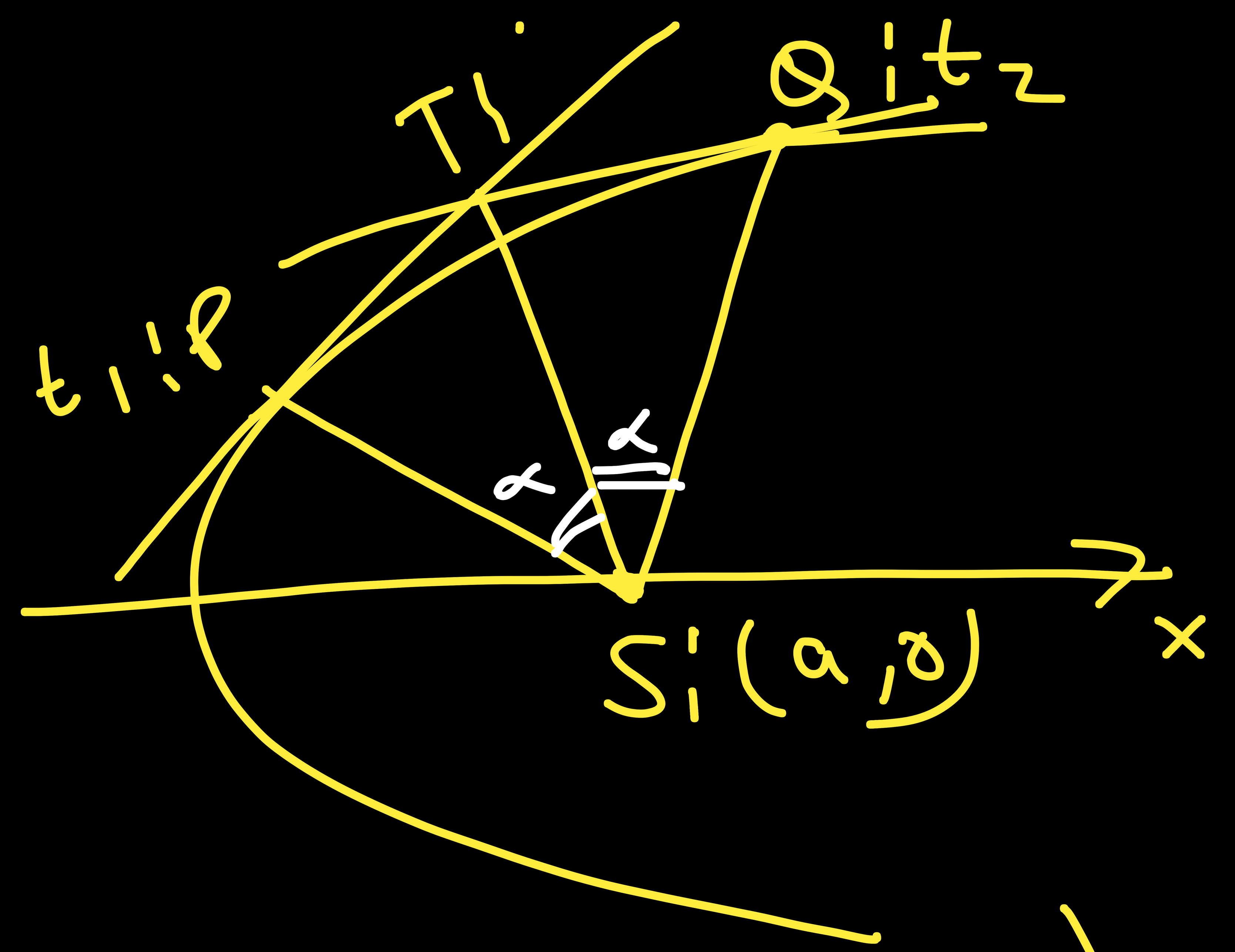
$$p_1 = a\,\frac{\left|\cancel{2}t_1^2 t_2 - t_1^3 + t_1 - \cancel{t_1^2 t_2} + t_2 - 2t_1\right|}{\sqrt{(t_1^2+1)^2}}$$

$$= a\,\frac{\left|t_1^2 t_2 - t_1^3 + t_2 - t_1\right|}{t_1^2+1} = \frac{a\left|t_1^2(t_2-t_1) + 1\cdot(t_2-t_1)\right|}{(t_1^2+1)}$$

$$= a\,\frac{\cancel{(t_1^2+1)}\,|t_2-t_1|}{\cancel{(t_1^2+1)}} = a|t_2-t_1|$$

llly

$p_2 = a|t_2 - t_1| \Rightarrow ST$ is angular bisector. H.P. (i)



T : $(at_1t_2,\ a(t_1+t_2))$

$$SP = a(1+t_1^2)$$
$$SQ = a(1+t_2^2)$$
$$SP \cdot SQ = a^2(1+t_1^2)(1+t_2^2)$$

$$ST^2 = (at_1t_2 - a)^2 + (a(t_1+t_2) - 0)^2$$
$$= a^2\left((t_1t_2-1)^2 + (t_1+t_2)^2\right)$$
$$= a^2(t_1^2t_2^2 + 1 + t_1^2 + t_2^2)$$
$$= a^2(1+t_1^2)(1+t_2^2)$$

$$\boxed{ST^2 = SP \cdot SQ}$$

H.P. (i)

(iii) $\dfrac{ST}{SP} = \dfrac{SQ}{ST}$ & $\alpha = \alpha$

$\Downarrow$

$\Delta SPT \sim STQ$

H.P. (iii)

One

If l, m are variable real numbers such that $3l^2 + 4m^2 - lm - 3m = 0$, then variable line $lx + my = 1$ always touches a fixed parabola, [redacted]

1. Vertex of the parbola is

A) $\left(\frac{1}{3}, -\frac{4}{3}\right)$ B) $\left(-\frac{1}{3}, \frac{4}{3}\right)$

C) $\left(\frac{4}{3}, -\frac{1}{3}\right)$ D) $\left(-\frac{4}{3}, \frac{1}{3}\right)$

2. Focus of the parabola is

A) $\left(\frac{1}{3}, -\frac{1}{3}\right)$ B) $\left(-\frac{1}{3}, \frac{7}{3}\right)$

C) $\left(-\frac{1}{3}, \frac{1}{3}\right)$ D) $\left(-\frac{7}{3}, \frac{1}{3}\right)$

3. Directrix of the parabola is

A) $3y + 7 = 0$ B) $3y + 4 = 0$

C) $3y - 4 = 0$ D) $3y - 7 = 0$

$$3l^2 + 4m^2 - lm - 3m = 0$$

$$lx + my = 1$$

$$l = \frac{1 - my}{x}$$

1 $\longrightarrow$ B

2 $\longrightarrow$ C

3 $\longrightarrow$ D

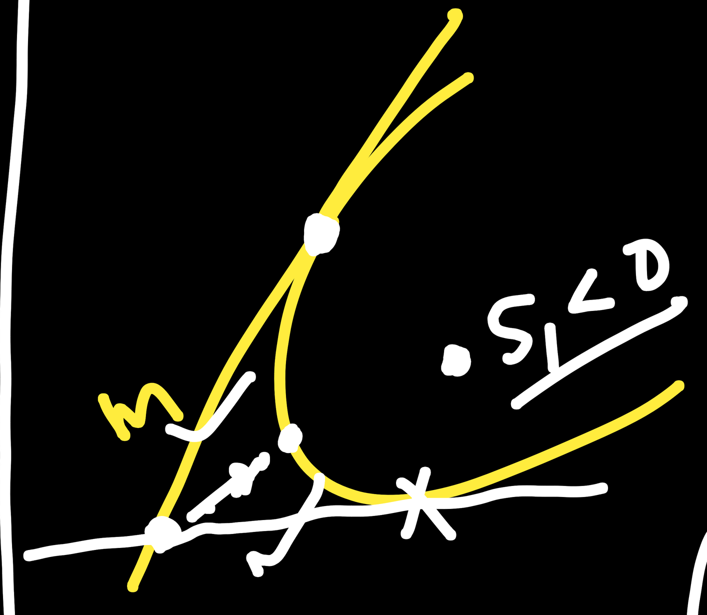
Concept of Envelope

$$y = mx + \frac{a}{m}$$

$\downarrow (h, k)$

$$k = mh + \frac{a}{m}$$

$$hm^2 - km + a = 0$$



$D > 0$

$k^2 - 4ah > 0$

$S_1 > 0$

$\boxed{D = 0}$

$$3l^2+4m^2-3m-lm=0$$

$$l=\frac{1-my}{x} \longrightarrow 3\left(\frac{1+m^2y^2-2my}{x^2}\right)+4m^2-3m+m\left(\frac{my-1}{x}\right)=0.$$

$$3+3m^2y^2-6my+4x^2m^2-3x^2m+xym^2-mx=0.$$

$$m^2(3y^2+4x^2+xy)-m(6y+3x^2+x)+3=0.$$

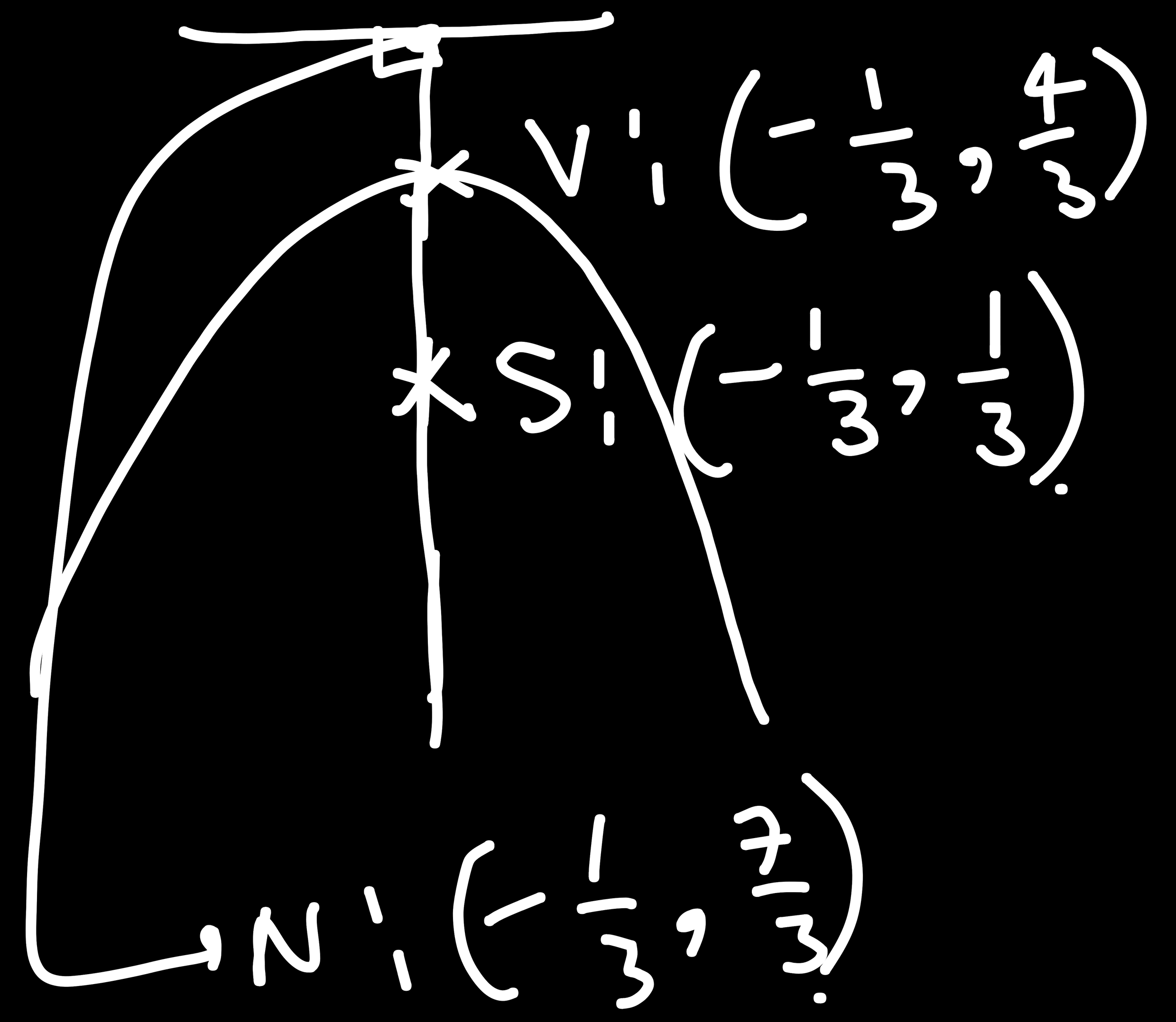
$$\Rightarrow D=0 \Rightarrow \cancel{36y^2}+9x^4+x^2+6x^3+\cancel{12xy}+36yx^2$$
$$-12(\cancel{3y^2}+4x^2+\cancel{xy})=0$$

$$9x^4+x^2+6x^3+36x^2y-48x^2=0 \qquad \div x^2$$

$$9x^2+1+6x+36y-48=0 \qquad \div 9$$

$$(3x+1)^2=-36\left(y-\frac{48}{36}\right)$$

$$\left(x+\frac{1}{3}\right)^2=-4\left(y-\frac{4}{3}\right).$$



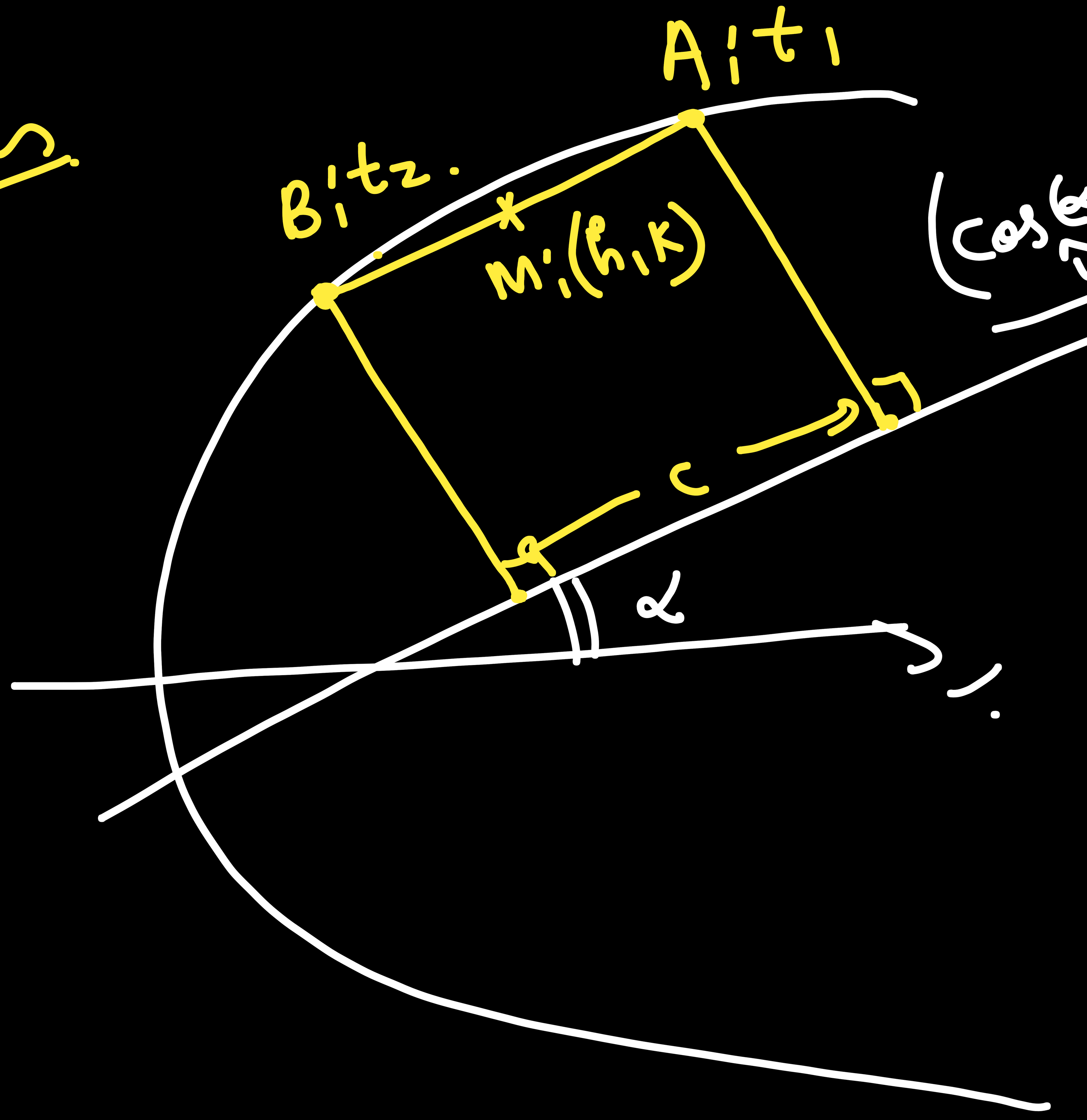
$y=7/3 \rightarrow$ Directrix

Que

A series of chords is drawn so that their projections on the straight line which is inclined at an angle α to the axis are of constant length c. Prove that the locus of their middle point is the curve

$$(y^2 - 4ax)(y\cos\alpha + 2a\sin\alpha)^2 + a^2c^2 = 0$$

Soln.



$$h = \frac{a(t_1^2+t_2^2)}{2}, \quad k = a(t_1+t_2)$$

$$k^2 = a^2(t_1^2+t_2^2+2t_1t_2)$$

$$k^2 = a(2h) + 2a^2 t_1 t_2.$$

$$(t_1-t_2)^2 = t_1^2+t_2^2-2t_1t_2.$$

$$\overline{AB} = a(t_1^2-t_2^2)\hat{i} + 2a(t_1-t_2)\hat{j}$$

$$|\overline{AB}\cdot\hat{u}| = c.$$

$$|a(t_1^2-t_2^2)\cdot\cos(\alpha) + 2a(t_1-t_2)\cdot\sin(\alpha)| = c.$$

$$a^2(t_1-t_2)^2\left((t_1+t_2)\cos(\alpha) + 2\sin(\alpha)\right)^2 = c^2.$$

$$a^2(t_1-t_2)^2\left((t_1+t_2)\cos(\alpha)+2\sin(\alpha)\right)^2=c^2$$

$$(t_1-t_2)^2=(t_1^2+t_2^2)-2t_1t_2$$

$$=\frac{2h}{a}-\cancel{2}\left(\frac{k^2-2ah}{\cancel{2}a^2}\right)$$

$$=\frac{2ah-k^2+2ah}{a^2}.$$

$$\cancel{a^2}\left(\frac{4ah-k^2}{\cancel{a^2}}\right)\left(\frac{k}{a}\cos(\alpha)+2\sin(\alpha)\right)^2=c^2$$

$$\Downarrow \quad \begin{array}{l} k\to y \\ h\to x \end{array}$$

$$(4ax-y^2)\left(y\cos(\alpha)+2a\sin(\alpha)\right)^2=a^2c^2$$

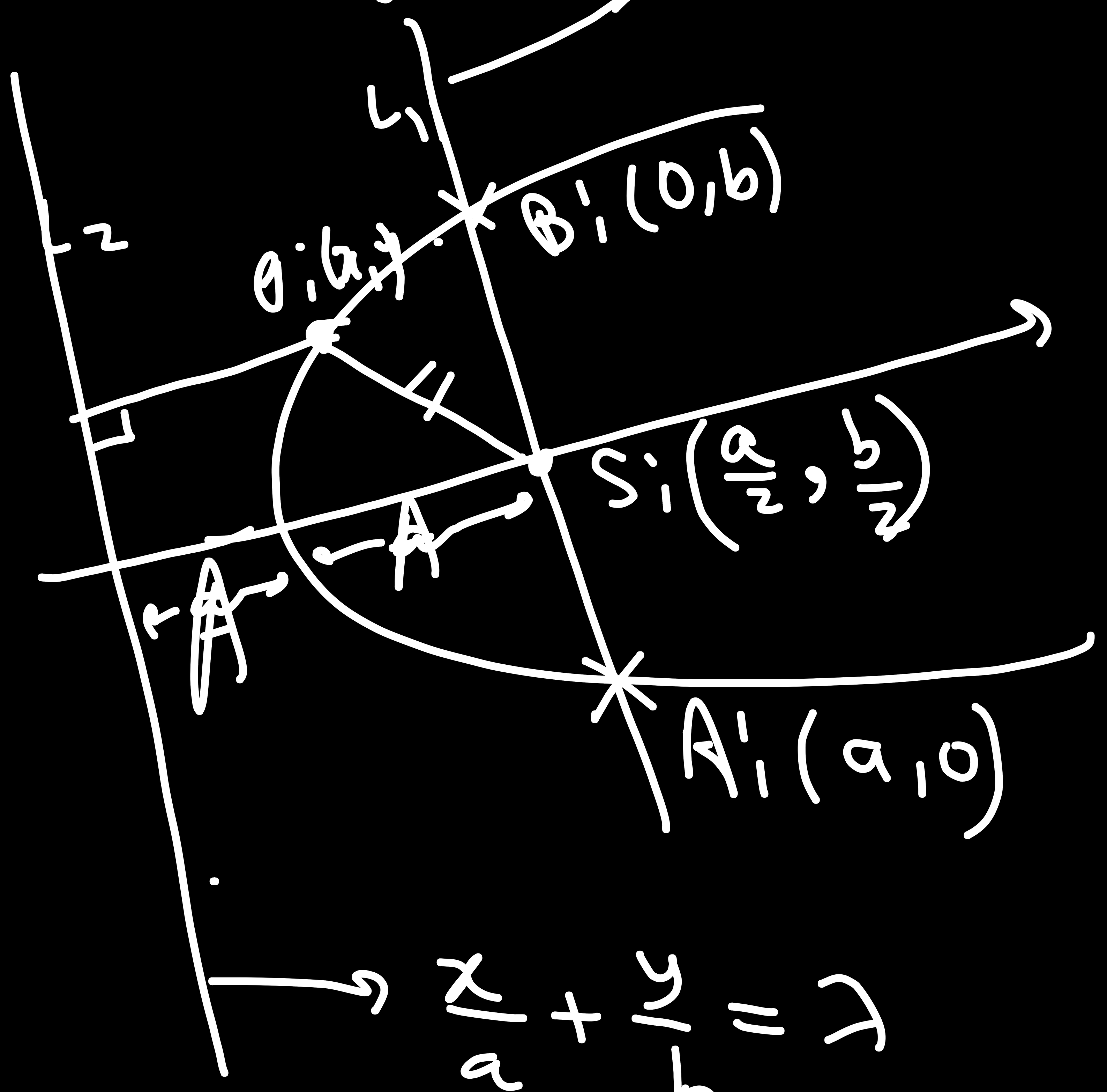
$$(y^2-4ax)\left(y\cos(\alpha)+2a\sin(\alpha)\right)^2+a^2c^2=0.$$

Que

A line AB makes intercepts of length a and b on the coordinate axes. Find the equation of the parabola passing through A, B and [redacted] if AB is the shortest focal chord of the parabola. Given A, B lie on X-axis & Y-axis resp.

Shortest focal chord $\Rightarrow$ AB is L.R.

$AB \equiv \frac{x}{a} + \frac{y}{b} = 1.$



$4A = \sqrt{a^2 + b^2}$

$2A = \frac{\sqrt{a^2+b^2}}{2}$

Distance b/w L_1 & L_2

$$\frac{|\lambda - 1|}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2}}} = \frac{\sqrt{a^2+b^2}}{2} \Rightarrow \lambda = 1 \pm \frac{a^2+b^2}{2ab}$$

$$= \pm\left(\frac{a^2+b^2}{2ab} \pm 1\right)$$

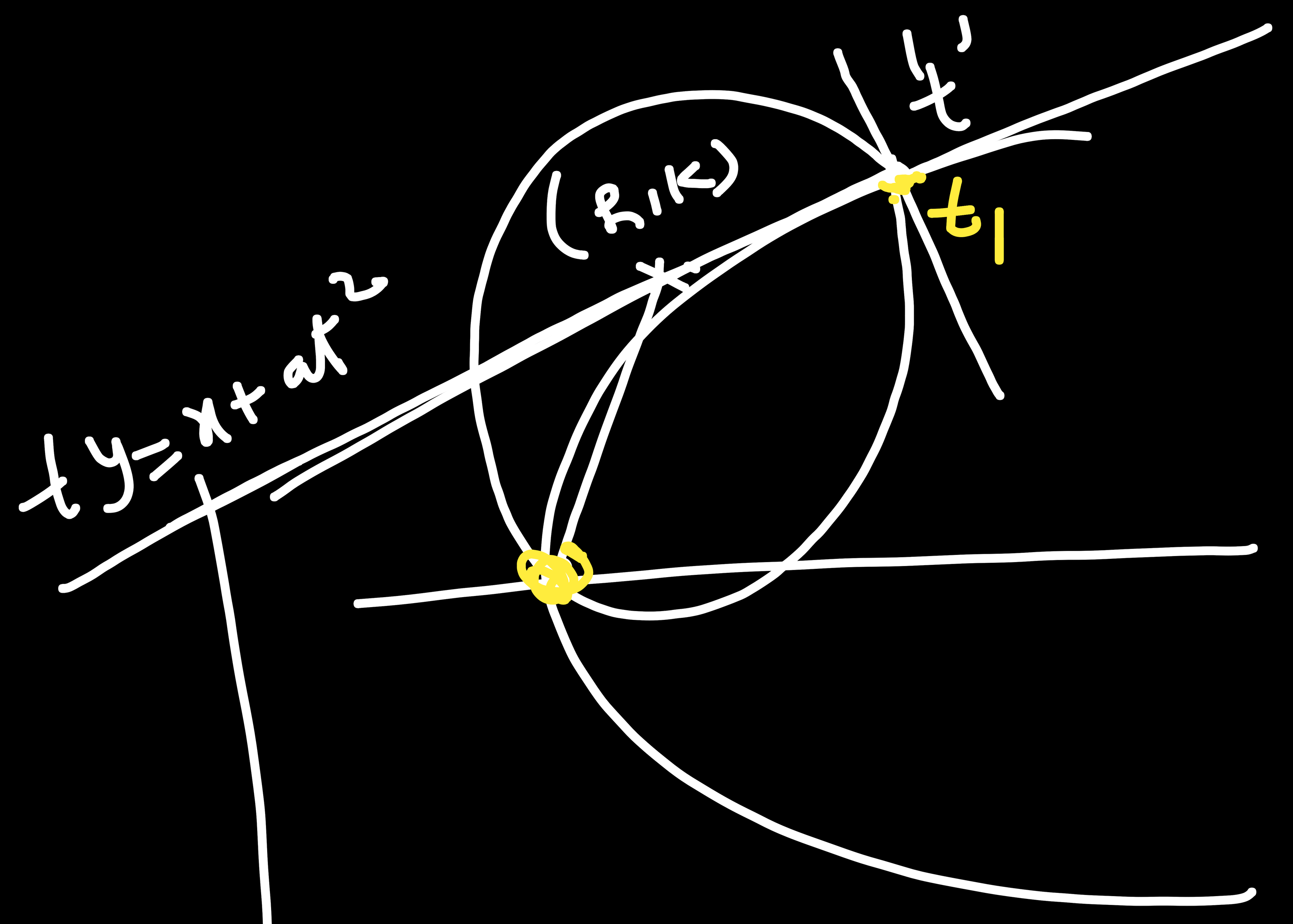
$$= \pm \frac{(a \pm b)^2}{2ab}$$

$PS^2 = PL^2$

$$\boxed{\left(x - \frac{a}{2}\right)^2 + \left(y - \frac{b}{2}\right)^2 = \frac{\left(\frac{x}{a} + \frac{y}{b} \mp \frac{(a \pm b)^2}{2ab}\right)^2}{\frac{1}{a^2} + \frac{1}{b^2}}}$$

Que

Find the locus of centres of a family of circles passing through the vertex of the parabola $y^2 = 4ax$ and cutting the parabola orthogonally at the other point of intersection.



$(x-h)^2+(y-k)^2=\left(\sqrt{h^2+k^2}\right)^2$

$x^2+y^2-2hx-2ky=0.$

$(at^2, 2at) \rightarrow a^2t^4+4a^2t^2-2hat^2-4akt=0.$

$\hookrightarrow at^3+(4-2h)t-4k=0 \rightarrow t_1, \ldots$ — (2)

$tk=h+at^2$

$at^2-kt+h=0 \rightarrow t_1$ — (1)

$\rightarrow at^3-kt^2+ht$

(2) $-t\cdot$(1)

$(4-2h)t-4k+kt^2-ht=0.$

$kt^2+(4-3h)t-4k=0$ — (3)

$$at^2 - kt + h = 0 \quad \text{——} \ 1$$

$$kt^2 + (4-3h)t - 4k = 0 \quad \text{——} \ 3$$

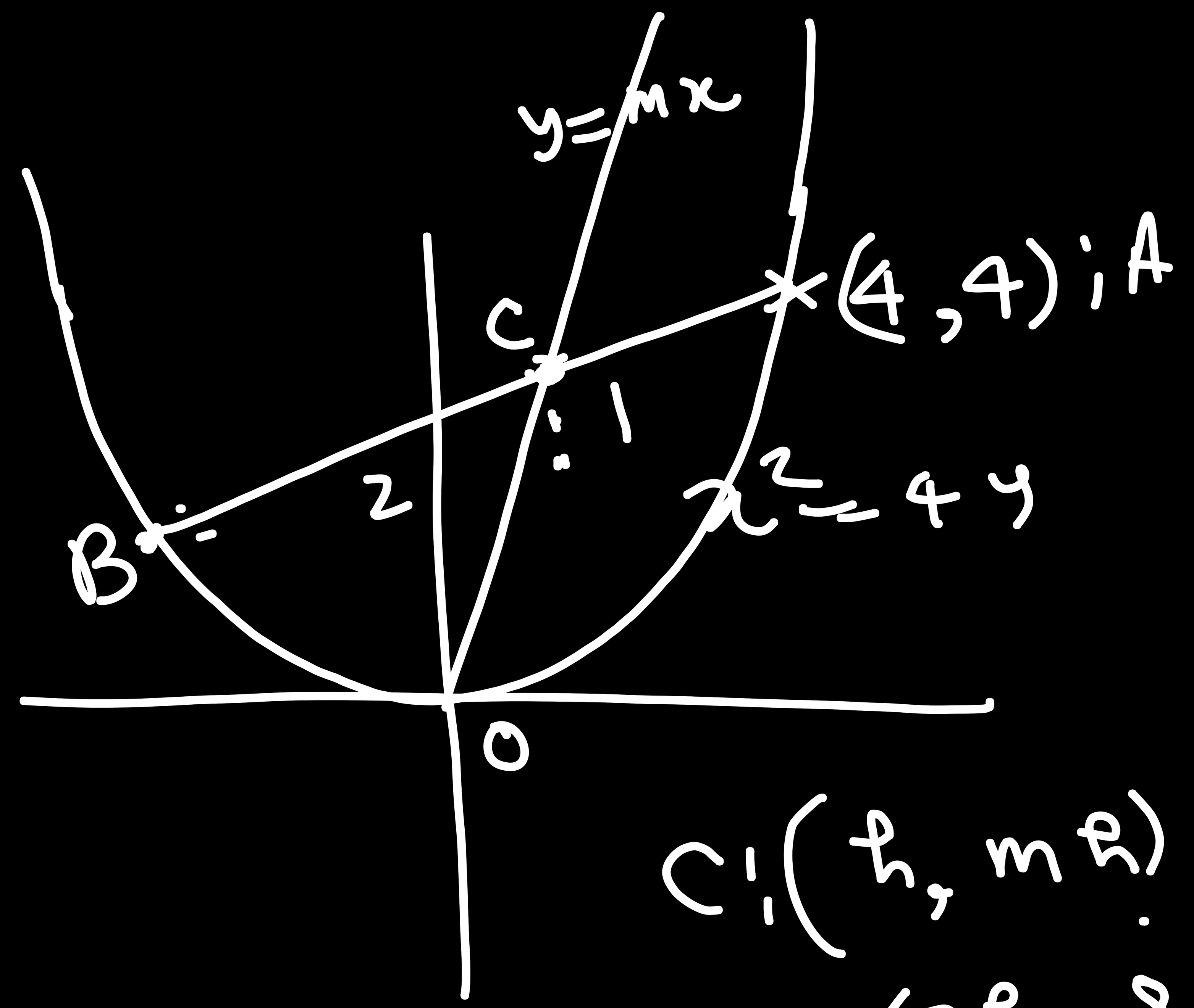
C.R.C.

$$(hk + 4ak)^2 = \left(a(4-3h) + k^2\right)\left(4k^2 - h(4-3h)\right)$$

$$(h, k) \longrightarrow (x, y)$$

$$\boxed{y^2(x+4a)^2 = \left(y^2 + 4a - 3ax\right)\left(4y^2 - x(4-3x)\right)}$$

If two chords drawn from the point (4, 4) to the parabola $x^2 = 4y$ are divided by line $y = mx$ in the ratio 1 : 2 then find the interval in which m lies.



$C : (h, mh)$

$B : \left(\frac{3h-8}{3-2}, \frac{3mh-8}{3-2}\right)$

$= (3h-8, 3mh-8)$

$x^2 = 4y$

$(3h-8)^2 = 4(3mh-8)$

$9h^2 + 64 - 48h = 12mh - 32.$

$9h^2 - (48+12m)h + 96 = 0.$

$D > 0$

$\cancel{12}^2 (m+4)^2 - \cancel{4} \cdot \cancel{9} \cdot 96 > 0$

19×4

$(m+4)^2 > 19$

$|m+4| > \sqrt{19}$

$\Rightarrow m \in (-\infty, -4-\sqrt{19}) \cup (-4+\sqrt{19}, \infty)$